Syam Son Penumala

Bangalore, India +91-73961-37983 syamson132000@gmail.com LinkedIn

Profile Summary

Motor Design Engineer with experience in BLDC/PMSM motor design, electromagnetic analysis, and motor optimization using Ansys Maxwell, Ansys Motor-CAD, and JMAG. Experienced in developing motors for washing machine, mixer grinder, drone & robotics, and VRF HVAC applications. Skilled in electromagnetic design, cogging torque reduction, thermal analysis, and prototype support.

Experience

R&D Engineer - Motor Design Aug 2025 – Present
PICL India Private Limited

- Working on Ansys Maxwell and Ansys Motor-CAD for electromagnetic design and analysis.
- End-to-end design and development of BLDC motors for home appliance applications.
- Conducting motor performance evaluation, optimization, and simulation studies.
- Supporting prototype development, testing, and validation activities.

Electrical Engineer May 2024 – Jul 2025
Vector Technics Pvt Ltd

- Designed and optimized BLDC motors using JMAG simulation software.
- Performed electromagnetic analysis, thermal evaluation, and torque-speed validation.
- Worked on motor development for drone and robotics applications.

Service Engineer Jul 2023 – Apr 2024
Vision Systems and Solutions

- Worked on Hyderabad Metro baggage scanner installation and commissioning.
- Conducted installation, maintenance, and troubleshooting activities.

Apprentice Mar 2023 – Jun 2023
Isuzu Motors India Pvt Ltd

- Worked in spot welding and paint shop operations.
- Gained exposure to manufacturing and quality control processes.

Education

Sagi Rama Krishnam Raju Engineering College Aug 2019 – Apr 2023
B.Tech in Electrical and Electronics Engineering

APSWR Junior College Jun 2017 – Apr 2019
Intermediate - MPC

St. John's EM High School Jun 2016 – Apr 2017
SSC

Projects

Power Generation Through Vertical Axis Wind Turbine

- Designed and developed a functional prototype of a Vertical Axis Wind Turbine (VAWT).
- Installed the turbine on top of the college building for real-time testing.
- Successfully generated electrical power to illuminate an LED bulb.

BLDC Motor Design for Drone and Robotics Applications

- Designed and developed a BLDC motor for drone and robotics applications.
- Performed electromagnetic design and optimization using JMAG.
- Evaluated torque-speed characteristics, efficiency, losses, and magnetic performance.

BLDC Motor Design for Washing Machine Applications

- Designed and developed BLDC motors for washing machine applications using Ansys Maxwell and Ansys Motor-CAD.
- Performed electromagnetic analysis, efficiency optimization, and thermal evaluation.
- Analyzed torque characteristics, losses, and overall motor performance.

BLDC Motor Design for Mixer Grinder Applications

- Designed high-speed BLDC motors for mixer grinder applications.
- Conducted electromagnetic simulations and performance analysis.
- Optimized motor efficiency, torque output, and compact design.

PMSM/BLDC Motor Design for VRF HVAC Applications

- Designed and optimized motors for Variable Refrigerant Flow (VRF) HVAC systems using Ansys Maxwell and Ansys Motor-CAD.
- Focused on cogging torque reduction through electromagnetic and geometric optimization.
- Conducted parametric studies on stator and rotor configurations.
- Evaluated back-EMF, torque ripple, losses, efficiency, and thermal performance.

Technical Skills

Simulation Tools: Ansys Maxwell, Ansys Motor-CAD, JMAG

Motor Design & Analysis: BLDC Motor Design, PMSM Design, Electromagnetic Design, Electromagnetic Analysis, Motor Optimization, Cogging Torque Reduction, Torque Ripple Analysis, Back-EMF Analysis, Thermal Analysis, Performance Validation

Engineering Domains: Home Appliance Motors, Drone Motors, Robotics Motors, HVAC Motors, Renewable Energy Systems

Software: MS Excel, MS Word, PowerPoint, Basic Photoshop

Professional Skills: Problem Solving, Analytical Thinking, Team Collaboration, Communication Skills